

Project Type: Engineering

Overall Grade: 76

Category Scores

Context Aims Objectives (Weight 2)

Score: 80

Justification: The report clearly defines the background, significance, and scope of the project. The aims are explicitly stated, with a set of numbered objectives that are referenced throughout the document. Some minor ambiguity exists regarding prioritisation, but overall the context is strong and demonstrates a good understanding of the problem space.

Transparency: Scoring was based on the explicit introduction of aims and objectives, strong contextual motivation (cybersecurity relevance), and consistent alignment with project outcomes, referencing the rubric criteria.

Reasoning Summary: The project establishes a solid foundation and clear direction, linking the work to the broader cybersecurity context.

Confidence: 95 – High clarity and explicit content in the aims and objectives provide strong evidence for confident scoring.

Literature Review (Weight 2)

Score: 77

Justification: The literature review surveys relevant research on data exfiltration and cybersecurity, referencing contemporary sources and distinguishing between existing, theoretical, and new methods. A good breadth of literature is demonstrated, though some sections could benefit from deeper critical analysis and synthesis linking literature more directly to the project's methodology.

Transparency: Score reflects the comprehensive overview of literature, variety of cited sources, and topical relevance per rubric, with some marks lost for limited critical evaluation.

Reasoning Summary: Relevant research is identified and contextualised, enabling the project to build upon and critique existing work.

Confidence: 90 – Substantial evidence of relevant literature, though minor gaps in critical discussion reduce confidence slightly.

Methodological Approach (Weight 1)

Score: 73

Justification: Methodologies and life cycles are appropriately covered, with justification for the incremental model based on project needs. There is discussion on software architectural decisions. The approach is logical and aligns with the aims, although some technical decisions could be expanded upon in terms of alternative options and reflections on their limitations.

Transparency: Scoring based on clear explanation and justification of development models and architecture, matching rubric expectations, with slight markdown for limited exploration of alternatives.

Reasoning Summary: Appropriate selection and justification of methodology support the structure and execution of the engineering project.

Confidence: 88 – The methodology is well-justified and described, with clear rationale for model and architecture choices.

Requirements Specification (Weight 3)

Score: 78

Justification: Requirements are well specified, divided into overall and functional categories, and derived logically from initial analysis and literature. Traceability is evident throughout design and implementation. Some requirements, such as usability considerations, could be elaborated further, but coverage is comprehensive.

Transparency: The score reflects the thorough enumeration and explanation of both functional and non-functional requirements, as dictated by the rubric.

Reasoning Summary: Detailed and appropriate requirements enable effective system design and assessment of project success.

Confidence: 92 – Clearly documented and well-structured requirements boost confidence in this assessment.

IT Design Analysis (Weight 3)

Score: 74

Justification: Detailed descriptions of design choices for various data exfiltration methods (content, header, meta) and supporting systems are given, with logical structuring and rationale. The design is clearly linked to the requirements. Some sections, such as the video steganography and MAC address exfiltration, acknowledge limitations and failures, but alternative design options are not systematically explored.

Transparency: Score reflects comprehensive design documentation and clear justification of decisions, with slight reduction for limited discussion of design trade-offs.

Reasoning Summary: Designs are robust and well-explained, demonstrating an understanding of how requirements are transformed into technical solutions.

Confidence: 89 – Comprehensive design evidence is provided, with only minor omissions reducing confidence.

Implementation Discussion (Weight 3)

Score: 75

Justification: The implementation section extensively documents the process for each exfiltration method, including successes and encountered obstacles. There is critical evaluation of failed attempts (e.g., video steganography, MAC address) and the application of workarounds. Testing context and methodological challenges are openly discussed, but more detail on code structure or repository organisation would strengthen the documentation.

Transparency: The score is based on the breadth and depth of implementation detail, as per the rubric, weighed against the absence of in-depth code presentation.

Reasoning Summary: Implementation is carefully described and linked to both design and requirements, including critical self-evaluation.

Confidence: 91 – Strong and thorough implementation narrative, with ample evidence provided.

Verification Validation (Weight 1)

Score: 71

Justification: Verification and validation are addressed through logical testing, user input validation, and Snort-based evaluations. There is good evidence of methodical testing and use of test data. However, the discussion could be enhanced by introducing additional performance or scalability checks.

Transparency: Graded based on documented testing strategies, output comparison via file hashes, and IDS validation, reflecting rubric requirements.

Reasoning Summary: Systematic testing validates the implemented methods against requirements and practical detection scenarios.

Confidence: 85 – The confidence is high due to structured and transparent test procedures, albeit with some scope for deeper analysis.

Evaluation Against Requirements (Weight 2)

Score: 76

Justification: Chapter 9 thoroughly evaluates the project against previously defined requirements, reporting on both the extent of success and areas where requirements were only partially met due to real-world limitations. The link between testing results and requirements is clear, though further quantitative analysis or user evaluation could bolster this section.

Transparency: Score awarded based on direct comparison of outcomes to initial requirements, with a balanced discussion of successes and shortcomings.

Reasoning Summary: Evaluation is detailed and honest, effectively assessing how well the project delivers on its stated goals.

Confidence: 90 – Strong evidentiary basis and clear evaluation framework support a high confidence score.

Project Planning Management (Weight 1)

Score: 69

Justification: Project planning is documented, including timelines, iterative development processes, risk management, and contingency planning (e.g., COVID-19 impact). Version control and backup protocols are referenced. However, progress monitoring and adaptive planning in response to encountered issues could be evidenced more systematically.

Transparency: The grade is justified by the inclusion of project plans and risk logs, assessed against rubric criteria for planning and management.

Reasoning Summary: Project management is adequate, with some evidence of good planning and risk mitigation.

Confidence: 81 – Moderate-to-high confidence based on documented plans and risk assessments, though project adaptation evidence is less prominent.

Solution Attributes (Weight 5)

Score: 77

Justification: The final solution supports multiple exfiltration methods, thorough testing and evaluation, and effective countermeasures. Attributes such as flexibility, adaptability, and

modularity are implied through the use of Python and compartmentalised coding. Some solution limitations (e.g., unfinished video steganography) are reported, but overall robustness and utility are high for the project's intended audience.

Transparency: The score reflects the solution's breadth, adaptability, and technical completeness, as required by the rubric.

Reasoning Summary: The developed test bed is functional and versatile, supporting various data exfiltration scenarios and contributing meaningfully to cybersecurity practice.

Confidence: 93 – Comprehensive documentation of solution features and evaluation against solution quality criteria gives high confidence.

Summary Conclusions Recommendations (Weight 2)

Score: 78

Justification: The work concludes with a clear summary of findings, reflective insights on problems encountered, and actionable recommendations for future research and development. The section is balanced and constructive, demonstrating a mature level of critical thinking.

Transparency: Scoring is based on the presence of clear conclusions, self-reflection, and well-justified recommendations as per the rubric.

Reasoning Summary: The conclusion synthesises key findings and prescribes meaningful directions for ongoing work.

Confidence: 89 – Well-structured and reflective conclusion adds confidence to the assessment.

Structure Presentation (Weight 2)

Score: 75

Justification: The report is well-organised with a logical structure, clear sectioning, appropriate use of figures and tables, and professional formatting. Some minor typographical and grammatical inconsistencies are present, but overall readability and navigation are strong.

Transparency: Grade reflects the structured and accessible presentation, as outlined in the rubric, with deductions for minor formatting or language lapses.

Reasoning Summary: Comprehensive and coherent structure underpins clear communication of ideas and results.

Confidence: 86 – High confidence due to evident organisational clarity, reduced slightly by some minor errors.

Overall Understanding Reflection (Weight 3)

Score: 79

Justification: The project demonstrates a deep understanding of the field, integrating literature, practical implementation, and evaluation. The author acknowledges limitations, reflects on methodological choices, and discusses the broader cybersecurity implications. Reflection on both achievements and failures is candid and demonstrates mature self-assessment.

Transparency: Score is based on evidence of integrated understanding, critical self-reflection, and awareness of the work's wider context, in line with rubric criteria.

Reasoning Summary: The report reflects comprehensive domain knowledge and a balanced, critical perspective.

Confidence: 92 – The work demonstrates depth and maturity, with high assurance in the logical and reflective discussion.

Strengths

The report excels in its clear articulation of objectives, comprehensive literature review, robust requirements specification, multi-method implementation, and thorough testing. The evaluation against requirements is candid and well evidenced. The project demonstrates strong critical self-reflection, situating findings within the broader context of cybersecurity practice.

Weaknesses

Critical analysis in the literature review could be deeper, some methodological alternatives are not fully explored, and the solution could benefit from more extensive user or performance evaluation. Certain implementation sections lack detailed code or reproducibility information, and minor proofreading issues are present.

Rubric Alignment

Summary: The marking closely follows the Engineering rubric, ensuring all categories are evaluated from context through implementation to reflection. Each section is graded in accordance with the defined criteria and weighted for overall assessment. Minor areas of underdevelopment are reflected in slightly reduced scores.

Threshold Checks

Missing Criteria: No essential rubric elements are wholly absent. Some categories, such as alternative methodological consideration and deeper critical literature analysis, could be further strengthened, but core criteria are met.

Overall Confidence: 91 – Overall confidence is high, as category-level confidences are consistently strong and supported by explicit evidence throughout the report.